In single-variable calculus, we regularly encounter functions that arise via composition. Differentiating these require use of the *chain rule*. Specifically, given a single-variable function $w = f(x(t))$, we see that t is the independent variable and x is the 'intermediate' variable. When computing $\frac{dw}{dt}$, we get

$$\frac{dw}{dt} = \frac{dw}{dx}\frac{dx}{dt}$$

If we *had* to draw a picture to describe this process, we might draw the above:

Sometimes in multivariable there exists multiple 'paths' through this type of picture since a function like $w = f(x(t), y(t))$ has *two* 'intermediate' variables. Thus the picture we might draw is the following:

Now, reconsider the function $w = f(x(t), y(t))$. Technically, this is a real-valued function that only has *one* independent variable. Thus, it makes sense to compute the 'total derivative'

$$\frac{dw}{dt},$$

where we used 'd' and not '∂' since we didn't make any choices. But! We must account for the myriad of intermediate paths:

Theorem. If $w = f(x(t), y(t))$, $x(t)$, and $y(t)$ are all differentiable, then

$$\frac{dw}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt}.$$

Note that care needs to be taken when evaluating this derivative at some point t_0. To see this, consider the following example:

Example. Consider the function $w = xy$, where $x = \cos(t)$ and $y = \sin(t)$. Compute $w'(t)$ using the multivariable chain rule and then verify this conclusion by computing $\frac{d}{dt}[w] = \frac{d}{dt}[\cos(t)\sin(t)]$.

Of course, this approach to the chain rule generalizes nicely to higher dimensions. For example, given a trivariate function $w = f(x(t), y(t), z(t))$, the value for $\frac{dw}{dt}$ is given by:

Example. Suppose the function $w = T(x, y, z) = xy + z$ returns the temperature of a specific spot in a heating vent. Due to a fan at the bottom, however, you know that the temperature changes most frequently for points (x, y, z) that lie on an upward spiral (i.e., the 'helix'). This path is traced our when

$$x(t) = \cos(t), \qquad y(t) = \sin(t), \qquad \text{and} \qquad z(t) = t.$$

Thus, if you want to know how quickly the temperature is changing *for points on the helix*, then you want to compute $\frac{dT}{dt}$ and need to view x, y, and z as intermediate variables living on the helix. Use this find how fast the temperature is changing at $t = 0$.

Functions Defined on Surfaces

Let us continue this temperature idea. Suppose that the function $w = f(x, y, z)$ returns the temperature for points (x, y, z) on the Earth's surface. In this manner, however, each x, y, and z are now functions of, say, longitude and latitude values (call these r and s, respectively). This means

$$w = f(x(r, s), y(r, s), z(r, s)).$$

Now f itself is a bivariate function dependent on r and s. Thus, it is reasonable to ask for something like $\frac{\partial f}{\partial r}$. [Note: It makes no sense to ask for $\frac{df}{dr}$, since we had to choose r over s.] So how to we compute these partials? Let's draw some diagrams:

Example. If $w = x + 2y + z^2$, where, $x = \dfrac{r}{s}$, $y = r^2 + \ln(s)$, and $z = 2r$, then compute both partials of w.

Example. If $w = x^2 + y^2$, where $x = r - s$ and $y = r + s$, then compute both partials of w.